K.R. Mangalam University

School of Engineering and Technology

LAB 2 BENCHMARKING REPORT

Program: BCA (AI & DS)

Course Code - ENCA351 (2026-27)

Submitted by:

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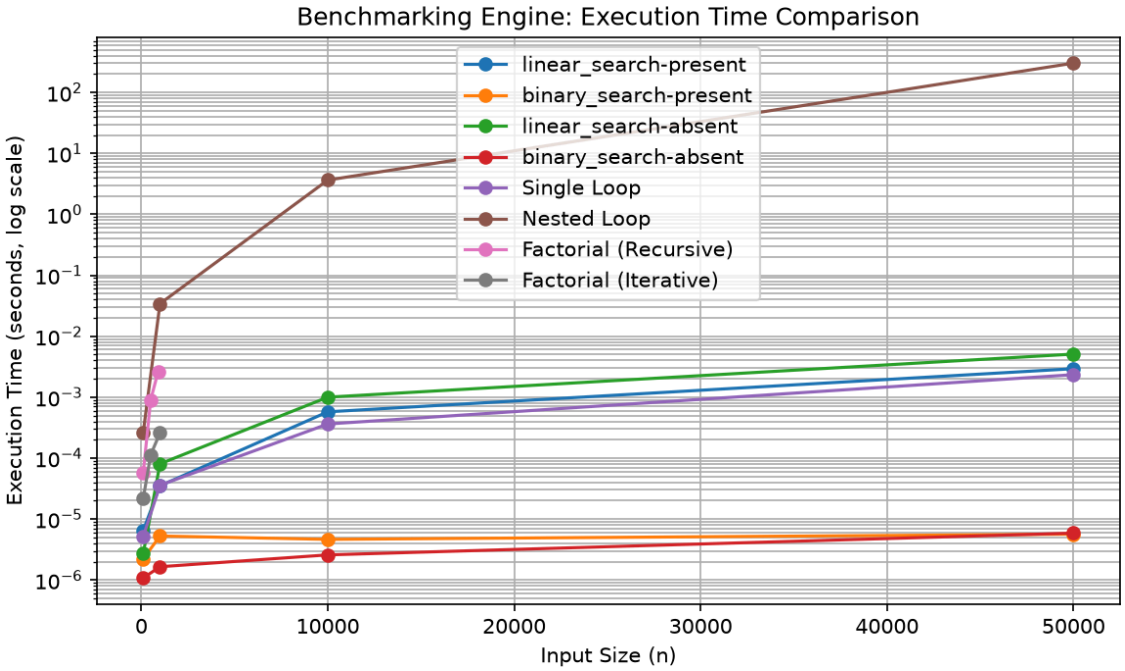
Course: BCA (AI & DS) - Section B

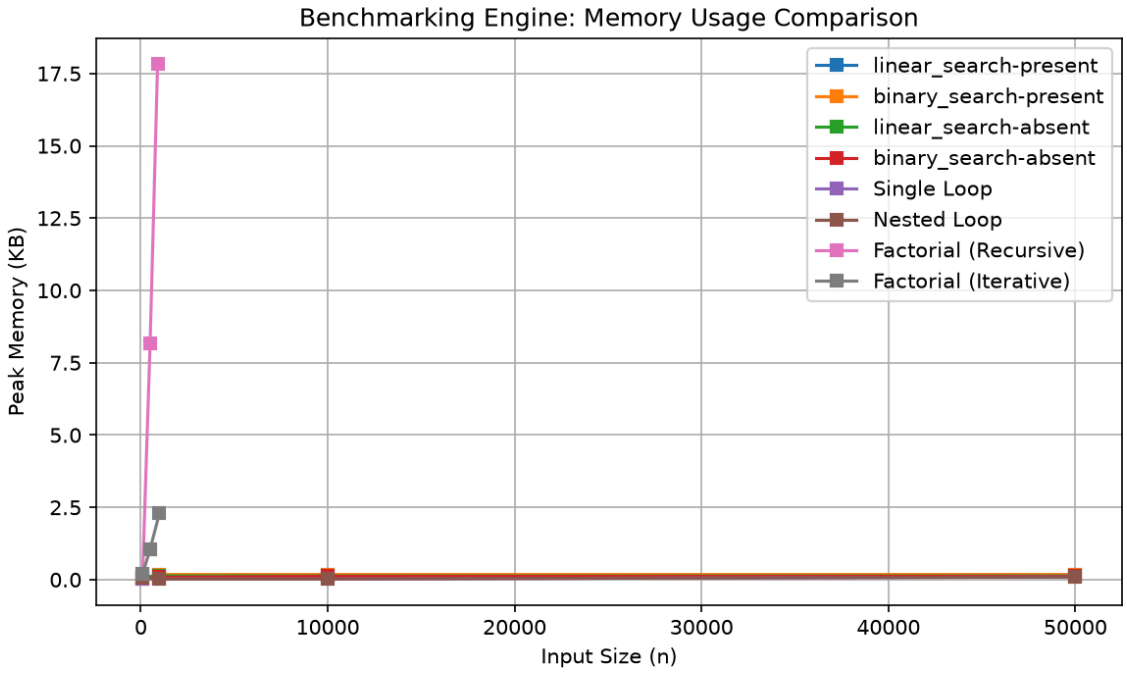
Submitted To: Dr. Aarti Sangwan Faculty, SOET

Graphs and Visualizations

Visualizations

Time axis is log-scale (10^-6 to 10^2 seconds); memory is linear. Nested Loop's curve leaves every other pattern behind, binary search stays flat, and recursive factorial's memory climbs while iterative stays level.





5. Analysis

Binary Search beats Linear everywhere measured (at 50k absent: 15 comparisons and 6 microseconds vs 50000 comparisons and 5 milliseconds, ~800x faster) because halving beats scanning; the price is sorted input. Recursive factorial loses to iterative on both axes (10x time, 8x memory at n=500) with identical results - stack frames cost real time and space, and the recursion ceiling caps n below 1000. Single vs nested loop is the sharpest lesson: same O(1) memory, but 50k inputs take 0.002s vs 302s, so nesting depth decides feasibility, not memory. Theory matches measurement throughout: linear counts track n, binary counts track log2(n) (6, 9, 13, 15), nested time tracks n^2.

6. Conclusion

Measure before deploying: binary search for sorted data, iterative factorial always, and never nest loops over large inputs. The engine, notebook, and app reuse the same measurement path, so every number in this report regenerates from one command.

7. How to reproduce

```
python search_analysis.py && python code_benchmark.py && python benchmark_engine.py
python run_full_sweep.py all && python visualize.py
```

Launch the interactive app with `streamlit run app.py`.

Executed live for this report:

- Module demos: binary 9 vs linear 501 comparisons at n=1000 present.
- `run_full_sweep.py search` plus `loops`: 30-row `data/sweep_full.csv`, nested-50k measured twice (302s, 283s).
- `visualize.py` plus full-data replot: log-scale `sweep_time.png`.
- `project_notebook.ipynb`: executes 3/3 cells, zero errors.
- Streamlit `app.py`: serves HTTP 200 headless.